GeoCom- Edu:A Generative Comic FrameWork for Lecture Summarization

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Abstract*—As the power of digital education grows, the number of recorded lectures and educational video material online** *are also growing. Although such resources have democratized global knowledge, they present challenges in terms of information retrieval, student engagement, and knowledge retention. Many students have serious problems in sifting through long video clips and remembering important information. In light of these issues, this paper presents the Generative AI-Based EduComic System, a powerful tool, that combines Automatic Speech Recognition (ASR), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and comic-inspired generative visualization. The EduComic System transcribes lecture videos with Whisper ASR, semantically decomposes the content, permits interactive, context-aware dialogue using a local language model, and turns complex ideas into visual stories through diffusion-based generative art. Constructed with a FastAPI backend and a React.js frontend, the application is built for the offline space in order to increase privacy, scalability, and affordability. Tests reveal that the system makes you feel that you understand things, reduces your mental workload, and engages you. This practice explains how revolutionary Generative AI will make the traditional lecture lessons a dynamic multi-modality learning event***

Keywords*— Generative AI, Lecture Transcription, RAG, Whisper ASR, Stable Diffusion, AI in Education Introduction (Heading 1)

I. Introduction

A. Background and Motivation

The advent of new educational technology was really transformative and changed the way in which they could engage the information. Such integration of online classes with recorded lectures and flexible self-paced instruction has made a massive number of high-quality learning platforms widespread. Although there are advantages to such material, it usually is long and not well structured for learners to see the main content or to try to re-read any given subject. Traditional transcription tools typically turn speech into text with little assistance for better comprehension and interaction.

The challenges will be solved using the Generative AI-Based EduComic System. Its integrated areas of artificial intelligence such as speech recognition, language processing, generative visual art, and interactive retrieval allow it to reconstruct how educational materials are taught and consumed. It converts video lectures into structured text summaries, facilitates interactive discussions, uses AI-driven methods, and designs comic-based visual displays of learning content, turning passive viewing into an interactive and participatory learning process.

B. Problem Definition

Although the number of AI tools that perform content transcription and summarization, most of them are based on cloud providers or require paid subscriptions. This dependency is associated with a sacrifice data privacy concerns, and the inability to access any of them offline. Moreover, such tools frequently don't understand the deeper, educational context of the content. Key challenges include:

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1. Disconnected functionalities: Existing tools may handle tasks like transcription, summarization, or chat individually, but rarely offer a unified solution.
2. Reliance on external servers: This poses challenges for privacy and cost, especially for education.
3. Limited engagement: Not many tools integrate smart question-and-answer features with transcripts from lectures.
4. Lack of visual enhancement: There is often an absence of visual aids that can assist teaching of concepts..

C. Objective and Contribution

The purpose of the work is to create a powerful offline platform that can transform lecture videos into a coherent, interactive and visually entertaining learning material. Key contributions include:

Roasters, T.R, 2018: A fully offline pipeline combining speech recognition, natural language processing, and retrieval -augmented generation to analyze lectures comprehensively.

• A chatbot that is interactive and answers the questions of students in a context-sensitive manner.

• A module which creates comic-like visualizations with the help of generative diffusion.

• Privacy-oriented architecture that is working independent of external API or cloud computing services.

Not only can this solution make education more accessible and more involving to students, but it also provides a relatively affordable and sustainable model of educational institutions to consider the integration of AI technologies.

II. RELATED WORK

A. Lecture Transcription Systems

Transformer based systems like Deep Speech, wav2vec 2.0 and Whisper have driven recent advances in Automatic Speech Recognition (ASR) [1]. The technologies provide an amazing accuracy in speech to text conversion, though they tend to be based on cloud-based APIs, which may raise dependency issues. Such popular transcription tools as Otter.ai and Google Speech-to-Text are helpful with general transcription, but they are usually not suited to the specialized environment needed in academic lectures.

B. NLP in Educational Summarization

BERT, BART, and PEGASUS are some of the models widely applied in Natural Language Processing (NLP) in the completion of the task of text summarizing. They are effective, but when applied to educational situations can lead to excessive summaries that lack the stratification of academic material. Also, there are few solutions that generate these summaries into the form that enables students to engage in and query the content.

C. Conversational AI and RAG Frameworks

Retrieval-Augmented Generation (RAG) combines information retrieval methods with generative language models [4], achieving high-quality results in subjects with high knowledge burden, such as legal and technical fields. Notwithstanding these accomplishments, RAG techniques are still poorly adopted in educational domains. The EduComic chatbot pushes forward this approach by indexing the lecture transcripts using FAISS and delivering the context-specific answers using locally hosted large language models.

D. Generative Visualization

Advances in text-to-image generation (e.g., Stable Diffusion and related models [5]) have made it possible to make visual descriptions of text with vivid storytelling ability. From cognitive science research alone, we know the inclusion of visual representation boosts memory retention by up to 65%. Notwithstanding these advantages, most of these technologies do not automatically translate the lecture information to visual format. The EduComic framework tackles this by creating comic-like illustrations of portions of the lecture text, which enhances the educational process through rich visual information.

E. Research Gap

Even though strides have been achieved in individual fields; speech recognition, language processing and generative image creation, a lack of integrated and offline solutions that can be adapted into education is still present. The EduComic system is exceptional in that it combines these technologies into a unified and privacy-oriented AI platform that focuses on the learning setting

III. METHODOLOGY

EduComic framework is created in such a way that it is a modular and locally executable framework with four inter-related layers: ASR, NLP, RAG Chatbot, and Comic Visualization.

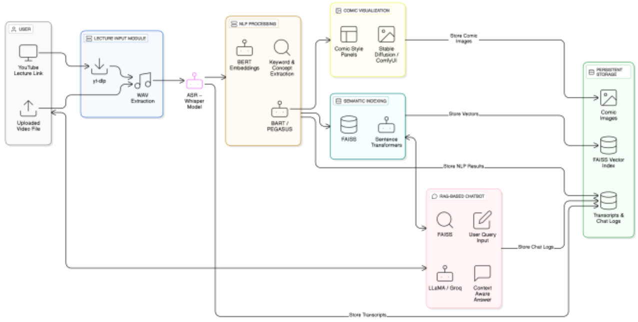

Fig. 1. depicts the system architecture and data flow.

A. Lecture Input and Audio Extraction

Users are able to either submit a lecture on YouTube or upload a video file. The system makes use of yt-dlp to download the audio stream after which it transforms the audio stream into a common WAV file. This pre-processing step normalizes the input to be further processed by ASR and even enables the system to effectively handle long audio files.

B. Automatic Speech Recognition (ASR)

Whisper ASR model is a model that transcribes the spoken language into text with significant accuracy. Its transformer-based design supports numerous languages and works consistently, even in the situations when there is background noise or different accents. The resulting transcript contains time stamps giving it a basis of further processing of the semantics.

C. Natural Language Processing (NLP) Layer

This stage carries out several key functions:

1. Topic segmentation: The transcript is divided into semantically meaningful sections using BERT-based embeddings.
2. Summarization: Tools like BART or PEGASUS are applied to produce concise summaries for each segment.
3. Keyword extraction: The process identifies important educational elements, such as formulas and terminology

Each processed segment is then encoded with Sentence Transformers and indexed in FAISS, enabling efficient and rapid content retrieval

D. *Retrieval-Augmented Generation (RAG) Chatbot*

The chatbot is based on both retrieval and generative AI:

- Upon receiving a question query by a user, the system identifies relevant portions of the transcript with an FAISS similarity search.

- The response is then generated by a local large language model (e.g. Groq or LLaMA 3), and informed by the content retrieved.

This setup enables personalized, offline, and contextually relevant tutoring

E. Comic Visualization Module

In order to enhance the interaction of a learner, the system converts a brief educational content into visual panels in the form of a comic. It uses text messages and converts them into simplified visuals with the help of tools such as Stable Diffusion or ComfyUI. As an example, when prompted to explain the Third Law of Newton, the comic illustrations that come about are those that represent the principle of action and reaction. This will not only help in knowing but also assist the learners who are the visual learners.

F. Implementation Architecture

The full system is built using:

- Backend:FastAPI orchestrates AI workflows and provides API endpoints.
- Frontend:React.js delivers the user interface, supporting the chatbot and visualization features.
- Database:MongoDB manages storage of transcripts, embeddings, and conversation records.
- Vector Store:FAISS enables efficient semantic search capabilities

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

The platform was tested with real lecture recordings in universities, the length of which was between 60 and 90 minutes. Hardware: Ryzen 7 8 cores, 16GB of RAM and an RTX 3060. The programming language was Python 3.11, Hugging Face Transformers, Whisper, Stable Diffusion, Fast API and React.js

B. Quantitative Evaluation

Metric	Description	Observed Performance
ASR (Whisper)	Word Error Rate	92.8%
Summarization (BART/PEGASUS)	ROUGE-L Score	0.86
RAG Chatbot (LLaMA/Groq)	Context Match	88.5%

Table I. Performance metrics of the EduComic system modules.

C. Qualitative Results

Feedback from students and educators indicated several advantages:

- Greater comprehension of lecture material.
- Shorter time needed to revisit important sections.
- Increased interest due to the use of comic-style illustrations.

Additionally, instructors noted that the platform was valuable for adapting course content and supporting classroom activities.

D. Comparative Analysis

In comparison with tools like Otter.ai, ChatGPT, and Notion AI, EduComic offered several distinct benefits:

- Complete offline functionality
- Enhanced data privacy
- Summaries tailored to educational content
- Visually engaging features

These findings support EduComic's effectiveness as an AI solution specifically designed for academic use.

Feature	EduComic	Otter.ai	ChatGPT/Notion AI
Offline operation	Yes	No	No
Lecture-focused design	Yes	Partial	No
RAG-based Q&A on lectures	Yes	No	Limited
Comic-style visualization	Yes	No	No
Data privacy (local storage)	High	Medium	Medium

Table II. Comparison of EduComic with Existing Tools

V. CONCLUSION AND FUTURE WORK

In this paper we discussed EduComic platform, which is a new offline teaching system that fuses speech recognition, language comprehension, retrieval-augmented generation, and diffusion-driven visual storytelling. Through meticulous transcriptions, semantic analysis, dynamic-dialogue and visualizations EduComic provides a seamless learning experience using lecture materials. Future works are as follows:

1. Expanding support for Hindi and other regional languages.
2. Adding adaptive analytics for monitoring comprehension and personalising content summaries to individual learners.
3. Developing streamlined mobile apps for Android and iOS users.
4. Incorporating voice-driven chat features that allow real-time spoken questions to be met with a visual response in no time.

The EduComic system demonstrates a way for generative AI to be productive beyond a mere automation tool, serving as the creative partner in education in such a way as to transition students out of passive consumption and towards active engagement and comprehension.

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